# Day 2 /Task 1

## 1. Fetch–Execute Cycle

The **Fetch–Execute Cycle** is the process by which the **CPU executes every instruction**. It repeats continuously until the program ends.

**Steps:**

1. **Fetch** – CPU fetches the next instruction from memory (RAM).
2. **Decode** – The Control Unit (CU) interprets the instruction.
3. **Execute** – The CPU performs the required operation.
4. **Repeat** – The Program Counter (PC) moves to the next instruction.

**Key Components:**

- **Program Counter (PC):** Stores the address of the next instruction.
- **Control Unit (CU):** Decodes and controls execution.
- **Registers:** Temporarily store data and instructions.
- **ALU:** Performs arithmetic and logical operations.

**Flow:**

```
Fetch → Decode → Execute → Repeat
```

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## 2. Asynchronous Events

An **asynchronous event** is an event that occurs **unexpectedly while a program is running** (e.g., keyboard press, mouse click, timer, network message).

### Types

### Polling

- CPU continuously checks if an event has occurred.
- Simple but wastes CPU time.
- Suitable for simple systems.

### Interrupt

- A device sends a signal to the CPU when an event occurs.
- CPU temporarily pauses the current task, handles the event, then resumes.
- Efficient and faster than polling.

### Polling vs Interrupt

| Polling | Interrupt |
| --- | --- |
| CPU repeatedly checks for events. | Device notifies the CPU. |
| Wastes CPU time. | Efficient CPU usage. |
| Slower response. | Faster response. |
| Simple implementation. | More efficient but slightly complex. |

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## 3. Subroutine (Method in Java)

A **subroutine** is a **reusable block of code** that performs a specific task. In Java, it is called a **method**.

**How It Works:**

1. Program calls the method.
2. Method executes its code.
3. Control returns to the calling program.

**Advantages:**

- Code reusability
- Reduces duplication
- Easier debugging
- Better program organization
- Improves readability and maintenance

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## ⭐ Quick Revision (Exam Ready)

- **Fetch–Execute Cycle:** CPU executes instructions in the order **Fetch** → **Decode** → **Execute** → **Repeat**.

- **Asynchronous Event:** An unexpected event during program execution.
- **Polling:** CPU continuously checks for events.
- **Interrupt:** Device notifies the CPU only when an event occurs.
- **Subroutine (Method):** A reusable block of code used to perform a specific task and improve code organization.